Post-quantum transition for device bounded anonymous credentials: a EUDI wallet application

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Abstract

In 2026, member states of the European Union are expected to have running their own EUDI wallet applications instances compliant with the ARF architecture guidelines. The goal of the ARF architecture is to provide privacy-preserving anonymous credentials to the users, but the current implementation fails. In the meantime, in 2024 the European Union has published a roadmap for the post quantum digital transition of members digital infrastructures.

No current implementation of anonymous credentials is post quantum secure at the moment, with cryptologist and mathematician in top pubblications claiming that is not possible to build a post quantum anonymous credential scheme with the actual technologies, given hardware constraints and advanced requirements. So we asked ourselves: is it possible to build a post quantum device bounded anonymous credential scheme with the actual technologies, giving hardware constraints and technical difficulties? The answare is **yes**, it is possible.

Moreover, in this thesis we present a post quantum anonymous credentials model that can be deployed with the actual technologies and legacy infrastructures, with NIST standardized post quantum signature schemes. We went furhter and implemented the full model and run it into android devices in order to benchmark the performance and the results are surprising. We also present some optimizations that could be implemented in order to further improve the performance of the scheme, and we show how to implement them with the actual technologies.

Contents

1 Introduction

Anonymous credentials are a powerful tool for preserving privacy in digital interactions between users and service providers. Anonymous credentials were introduced by David Chaum in 1982 [6], and later formalized by Camenisch and Lysyanskaya in 2001 [3, 4]. They allow users to prove their identity or attributes without revealing any additional information about themselves.

The usual model of anonymous credentials consists of three parties:

- The issuer, a central entity who is responsible for issuing credentials to users and eventually revoking them;
- The user, who asks credentials to the issuer and holds them, and eventually use them to prove their identity or attributes to verifiers.
- The verifier, who is responsible for verifying the user's credentials without learning any additional information about the user.

A basic scheme of how this entity interacts with each other is represented in figure 1.

In the first approach [10], the issuer was signing user's credentials using a secure protocol that protects user's attributes. Credential(s) are then showed to a verifier via zero-knowledge proof of knowledge of a credential(s) whose attributes satisfy some predicate.

Systems based on anonymous credentials have been studied and adopted over the years by big tech companies like Micrsoft [11], IBM [5], and are in continuous

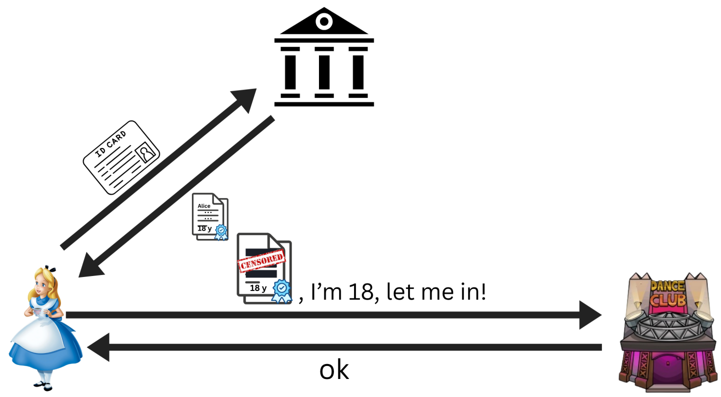

Figure 1: Basic scheme of anonymous credentials.

study and development by the scientific community, facing challenges like performance improvements, revocation lists, legacy infrastructures and hardware constraints, and more recently, post quantum security.

1.1 Post quantum transition

The european union on 11 April 2024 has published a roadmap for the post quantum digital transition of members digital infrastructures [9].

The roadmap is a comprehensive plan that outlines the steps that the European Union will take to transition to post-quantum cryptography. The roadmap includes a timeline for the transition, as well as a list of actions that will be taken to ensure a smooth transition.

The timeline is organized into three major phases:

1. **By the end of 2026**: Initial national PQC transition roadmaps have been established by all Member States, PQC transition planning and pilots for high- and medium-risk use cases have been initiated;

2. **By the end of 2030**: The PQC transition for high-risk use cases has been completed, PQC transition planning and pilots for medium-risk use cases have been completed. Quantum-safe software and firmware upgrades are enabled by default.

3. **By the end of 2035**: The PQC transition for medium-risk use cases has been completed. The PQC transition for low-risk use cases has been completed as much as feasible.

As previously said, this roadmap has been published by the European Commission in 2024. The recent advances in the field of post-quantum computing due to hardware improvements, artificial intelligence accelleration and advances in research have made the Q-day closer then expected, and the transition to post-quantum cryptography more urgent.

On 30 March 2026, Google published a drastical advance in post-quantum cryptography: they reduced consistently the amount of time, resources and qbits needed to break elliptic curves based crytography [2]. In the meanwhile, between April and March 2026, big tech companies like Google [1] and Cloudflare [12] have announced that they will accelerate the transition to post-quantum cryptography, lowering the timeline for the full transition to 2029, instead of 2035.

1.2 Overview

In this thesis, we present a post-quantum anonymous credentials model that can be deployed with the actual technologies and legacy protocols. We require the issuer to adopt ML-DSA NIST standardized post quantum signature scheme, and to sligthly leverage the credentials structure in order to accomodate the larger size of post quantum signatures and public keys, as well as modification to the mdoc structure in order to involve a commitment of the device public key instead of the device public key itself.

In the following sections, we are going to present an overview of the EUDI wallet 2, we perform a full ananlysis of the actual state of the art on device bounded anonymous credentials 3, then we explain how our solution faces and solves the challenges, performing an important advance on the actual state of the art 4. We show how we implemented our solution, giving a reference to our github repository 5, and we present our results 6, future works that we plan to do 7 and conclusion 8.

2 EUropean Digital Identity Wallet

In 2021, the European Commission proposed to adopt a framework for a European Digital Identity [7]. In 2024, the European Commission decided to establish an Architecture Reference Framework (ARF) [8], which consisted of a set of guidelines and properties that each instance of the wallet application should satisfy. This general framework is needed since the commission left freedom to eaech memeber state to implement their own instance of the application, due to different possible credentials structures (for example, french and german national identity cards have different structures and/or attributes) and infrastructures (for example, some member states have already a national identity card with an embedded chip, while others don't).

2.1 Architecture overview

2.1.1 Functionalities

2.1.2 Roles

2.2 Vulnerabilities

3 State of the art/Literture overview on device bounded anonymous credentials

3.1 Anonymous Credentials

3.2 Post quantum anonymous credentials

4 Our solution

5 Implementation

6 Results

7 Future work

8 Conclusion

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